

NQ/MNQ Range Alignment Backtest

9:00 AM 1H candle (BIAS) and 9:45 AM 5m candle (PRECISION)

Range level alignment vs continuation predictiveness

Symbol: MNQ continuous front-month (MNQ.c.0)

Source: Databento GLBX.MDP3, ohlcv-1m schema

Sessions: 253 valid trading days

Window: 2025-05-02 -> 2026-04-30

Default: lookback=5d, tolerance=5pts (NY tz, DST-aware)

Levels tested per candle: 0, 0.25, 0.50, 0.75, 1.00

(low wick, lower-quartile, midpoint/CE, upper-quartile, high wick)

An 'aligned' day = today's 9AM 1H level falls within tolerance of any prior-day's 9:45 5m level (within lookback window).

Outcome metric: post-10:00 close-to-close move and MFE/MAE in

5 windows: 10-11, 10-12, 10-13, 10-15, 10-16 (NY).

Primary post-10:00 window for headline stats: 10:00 - 13:00.

1. Headline summary (aligned vs non-aligned)

Primary window = 10:00 - 13:00 NY

group	count	avg_move	median_move	win_rate	continuation_rat	chop_rate	avg_chop_score	avg_MFE	avg_MAE
aligned	143	-0.62	25.50	0.52	0.78	0.01	0.24	100.88	-114.24
non_aligned	110	15.30	11.88	0.60	0.75	0.01	0.21	100.35	-98.56

2. Significance tests (two-prop z / Welch t)

None of the comparisons reach $p < 0.05$

	test	p1	n1	p2	n2	stat	p_value
aligned_vs_	nonaligned_contin	0.7832	143	0.7455	110	0.7041	0.4813
aligned_vs_	nonaligned_abs	115.2150	143	99.8977	110	1.2233	0.2212
scenario_	both_vs_none_con	0.7907	129	0.7262	84	1.0856	0.2777
scenario_	both_vs_only_1h_co	0.7907	129	0.8077	26	-0.1953	0.8451
scenario_	only_1h_vs_none_co	0.8077	26	0.7262	84	0.8337	0.4045
scenario_	only_5m_vs_none_co	0.7143	14	0.7262	84	-0.0923	0.9265

3. Scenario comparison

both = 1H+9:45 aligned, only_1h, only_5m, none

group	count	avg_move	median_move	win_rate	continuation_rat	chop_rate	avg_chop_score	avg_MFE	avg_MAE
both	129	-9.797	15.250	0.535	0.791	0.008	0.231	93.655	-116.517
only_1h	26	5.596	8.000	0.769	0.808	0.000	0.154	112.490	-107.500
only_5m	14	83.964	117.500	0.357	0.714	0.000	0.286	167.482	-93.214
none	84	18.307	13.125	0.548	0.726	0.012	0.231	96.598	-95.795

4. Tolerance sweep

Tighter tolerance (2pt) gives the cleanest cohort

group	count	avg_move	median_move	win_rate	continuation_ra	chop_rate	avg_chop_scor	avg_MFE	avg_MAE	tolerance_pts	tolerance_pct
aligned_2.0pt	100	2.0175	23.1250	0.5300	0.8000	0.0100	0.2300	94.2425	-108.1725	2.0000	
aligned_5.0pt	143	-0.6171	25.5000	0.5175	0.7832	0.0070	0.2364	100.8829	-114.2360	5.0000	
aligned_10.0pt	157	7.3185	29.7500	0.5541	0.7898	0.0064	0.2306	105.4729	-110.7818	10.0000	
aligned_0.025pt	151	5.5066	29.7500	0.5430	0.7947	0.0066	0.2291	105.6656	-112.6639		0.0003
aligned_0.050pt	158	8.7326	30.8750	0.5506	0.7911	0.0063	0.2316	106.4557	-110.9826		0.0005
aligned_0.100pt	164	6.1692	28.8750	0.5427	0.7927	0.0061	0.2305	104.7500	-111.7409		0.0010

5. Lookback sweep

1-day lookback outperforms 5/10/20

group	count	avg_move	median_move	win_rate	continuation_ra	chop_rate	avg_chop_score	avg_MFE	avg_MAE	lookback_days
lookback_1d	73	-20.938	5.250	0.562	0.808	0.000	0.247	93.301	-121.942	1
lookback_3d	126	2.292	27.375	0.532	0.786	0.008	0.233	99.377	-107.266	3
lookback_5d	143	-0.617	25.500	0.517	0.783	0.007	0.236	100.883	-114.236	5
lookback_10d	170	-5.868	14.875	0.541	0.788	0.006	0.234	96.910	-116.850	10
lookback_20d	186	-5.511	14.875	0.538	0.780	0.011	0.234	97.734	-117.663	20

6. Top level pairs by avg |move|

today_1h_level x prior_5m_level (lookback=5, tol=5)

today_1h_level	prior_5m_level	count	avg_move	median_move	abs_avg_move
1h_L100	5m_L25	10	-50.62	24.50	184.53
1h_L0	5m_L75	10	-107.95	-80.12	138.65
1h_L25	5m_L75	10	-23.25	-25.38	138.60
1h_L25	5m_L50	15	28.53	62.25	130.33
1h_L100	5m_L75	15	-43.70	-27.00	123.97
1h_L50	5m_L25	21	25.88	63.25	123.17
1h_L50	5m_L75	18	40.50	72.00	120.33
1h_L0	5m_L100	16	-28.38	-49.62	118.81
1h_L75	5m_L50	13	-43.42	-9.50	115.54
1h_L50	5m_L100	10	-34.83	8.25	110.42
1h_L100	5m_L0	14	11.46	18.00	107.00
1h_L75	5m_L0	14	-33.50	-14.62	103.93
1h_L100	5m_L100	22	-36.77	-15.00	100.64
1h_L100	5m_L50	13	-38.00	5.25	99.73
1h_L0	5m_L0	15	51.68	29.75	97.05
1h_L0	5m_L25	11	-31.16	-63.25	92.98
1h_L25	5m_L25	17	4.07	14.50	90.07
1h_L75	5m_L75	19	33.11	51.00	87.08
1h_L50	5m_L50	24	60.06	48.50	80.92
1h_L75	5m_L100	16	29.25	40.38	80.75
1h_L0	5m_L50	11	38.61	32.25	74.61
1h_L25	5m_L0	16	8.80	31.75	67.83
1h_L75	5m_L25	15	1.47	-14.25	67.47
1h_L25	5m_L100	14	10.07	14.50	66.07
1h_L50	5m_L0	17	-11.04	6.50	60.37

7a. Day of week

0=Mon, 1=Tue, 2=Wed, 3=Thu, 4=Fri

dow	count	avg_move	continuation_rate	chop_rate
0	52.000	25.938	0.788	0.000
1	52.000	23.452	0.731	0.019
2	52.000	3.370	0.788	0.000
3	50.000	-17.675	0.820	0.020
4	47.000	-5.633	0.702	0.000

7b. Range bucket (small/med/large 9AM range)

range_bucket	count	avg_move	continuation_rate
large	86.000	14.302	0.791
med	83.000	9.298	0.807
small	84.000	-4.842	0.702

7c. Per today_1h_level reaction

today_1h_level	count	avg_abs_move	avg_move	continuation_rate
1h_L0	63.00	104.55	-10.73	0.83
1h_L100	74.00	117.75	-31.14	0.82
1h_L25	72.00	95.59	7.59	0.76
1h_L50	90.00	98.06	24.20	0.78
1h_L75	77.00	89.81	1.11	0.74

7d. Per prior_5m_level reaction

prior_5m_level	count	avg_abs_move	avg_move	continuation_rate
5m_L0	76.00	85.79	5.52	0.78
5m_L100	78.00	95.34	-12.85	0.77
5m_L25	74.00	108.08	-2.90	0.80
5m_L50	76.00	98.90	16.26	0.80
5m_L75	72.00	117.40	-8.47	0.78

8. Cluster-size effect

More aligned levels per day -> higher continuation? (7+ hits = 100% on n=5)

n_hits	count	avg_move	continuation_rate	chop_rate
(-0.1, 0.0]	110.000	15.302	0.745	0.009
(0.0, 1.0]	46.000	-5.304	0.804	0.000
(1.0, 3.0]	57.000	13.127	0.754	0.018
(3.0, 6.0]	35.000	-18.357	0.771	0.000
(6.0, 100.0]	5.000	10.000	1.000	0.000

9. Hour-of-touch reversal (prior-zone touch test)

Tests prior 90% reversal claim at 13:00 -> NOT replicated

src	first_touch_hour	n	reversal_rate	avg_react_mfe	avg_react_mae
1h	10	1071	0.555	70.005	-82.894
1h	11	249	0.546	64.172	-89.369
1h	12	128	0.453	71.484	-67.400
1h	13	119	0.479	74.130	-73.634
1h	14	93	0.624	77.228	-59.336
1h	15	86	0.558	63.468	-47.645
5m	10	1149	0.560	69.251	-87.299
5m	11	233	0.554	59.723	-82.673
5m	12	106	0.528	97.434	-65.965
5m	13	126	0.468	73.861	-71.653
5m	14	85	0.541	75.388	-63.038
5m	15	98	0.541	67.421	-48.462

10. Spot-check dates (manual TradingView review)

	category	date	ten_open	post_10_13_mov	MFE	MAE	chop_score	continuation
best_alignment_win		2025-11-20	25275.25	-880.25	34.75	-967.00	0.40	continuation
best_alignment_win		2025-10-10	25357.25	-699.25	14.00	-709.50	0.40	reversal
best_alignment_win		2026-01-02	25733.75	-409.75	46.25	-439.75	0.00	reversal
best_alignment_win		2026-02-12	25208.25	-382.75	52.00	-402.75	0.00	continuation
best_alignment_win		2026-03-09	24371.25	337.75	378.75	-47.75	0.40	reversal
worst_alignment_fai		2025-11-10	25637.25	-2.25	29.25	-185.00	0.00	continuation
worst_alignment_fai		2026-03-12	24658.75	3.00	91.25	-97.75	0.00	continuation
worst_alignment_fai		2025-09-30	24788.00	-4.50	76.00	-41.75	0.40	reversal
worst_alignment_fai		2025-09-09	23816.00	5.25	34.75	-89.50	0.40	continuation
worst_alignment_fai		2026-01-06	25753.25	6.50	32.50	-114.25	0.40	continuation
best_nonalignment_t		2026-03-05	25185.25	-378.50	23.00	-400.00	0.40	continuation
best_nonalignment_t		2026-02-04	25196.75	-322.50	109.00	-412.75	0.00	continuation
best_nonalignment_t		2025-10-14	24575.25	316.75	372.00	-20.75	0.00	reversal
best_nonalignment_t		2025-11-14	24953.00	251.00	340.50	-90.50	0.00	continuation
best_nonalignment_t		2025-11-13	25382.00	-245.25	115.00	-256.25	0.40	continuation
choppiest_nonalignm		2025-08-01	22879.75	-0.50	165.75	-80.50	0.40	continuation
choppiest_nonalignm		2025-05-28	21467.00	-2.25	20.75	-40.75	0.00	continuation
choppiest_nonalignm		2025-06-11	21980.75	-3.00	86.00	-73.50	0.40	reversal
choppiest_nonalignm		2025-11-27	25296.25	3.50	11.50	-10.50	0.40	continuation
choppiest_nonalignm		2025-05-26	21279.75	4.75	16.50	-64.25	0.40	reversal

Chart 1 — Avg |move| aligned vs non-aligned

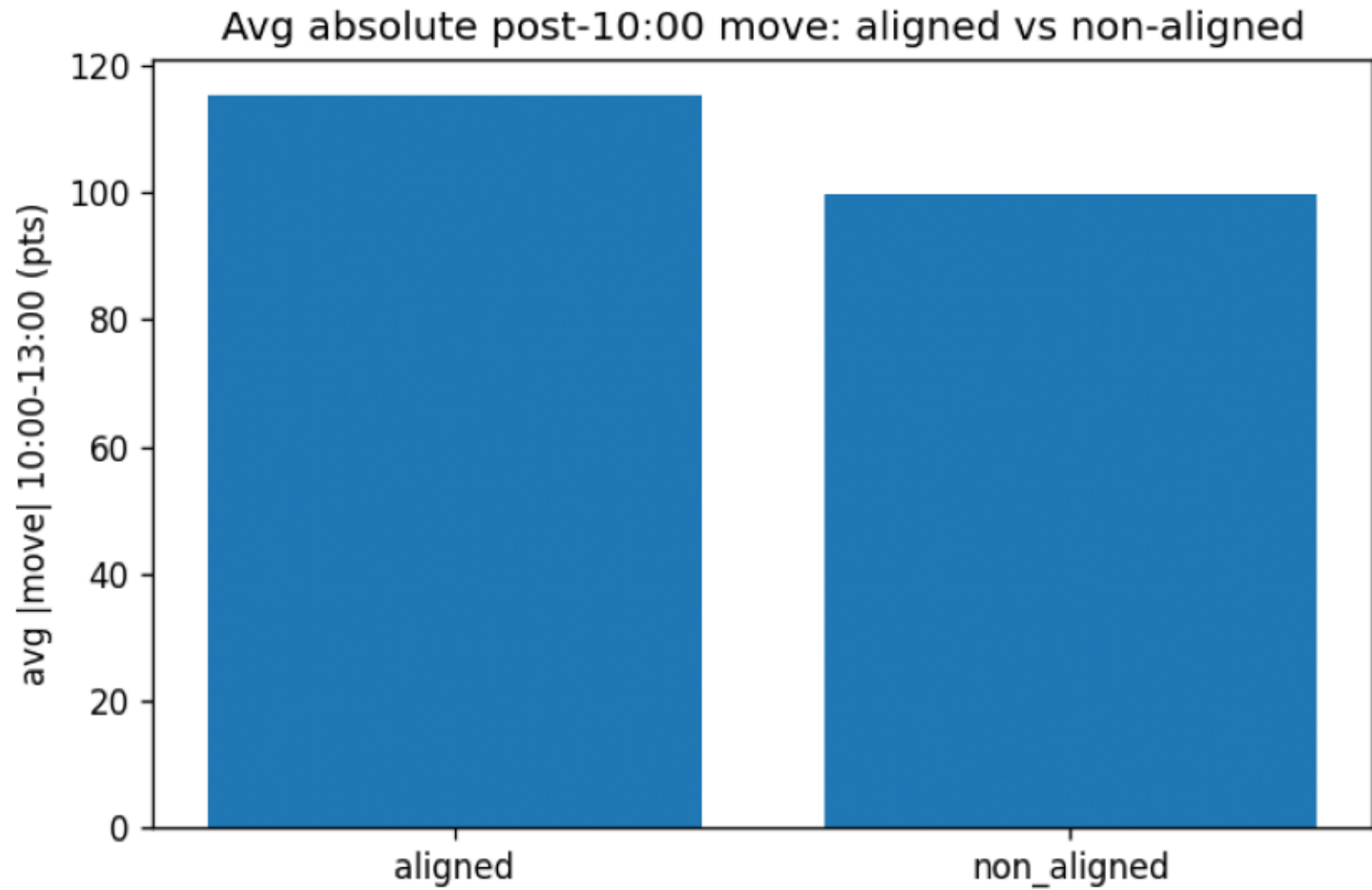


Chart 2 — Continuation rate vs tolerance

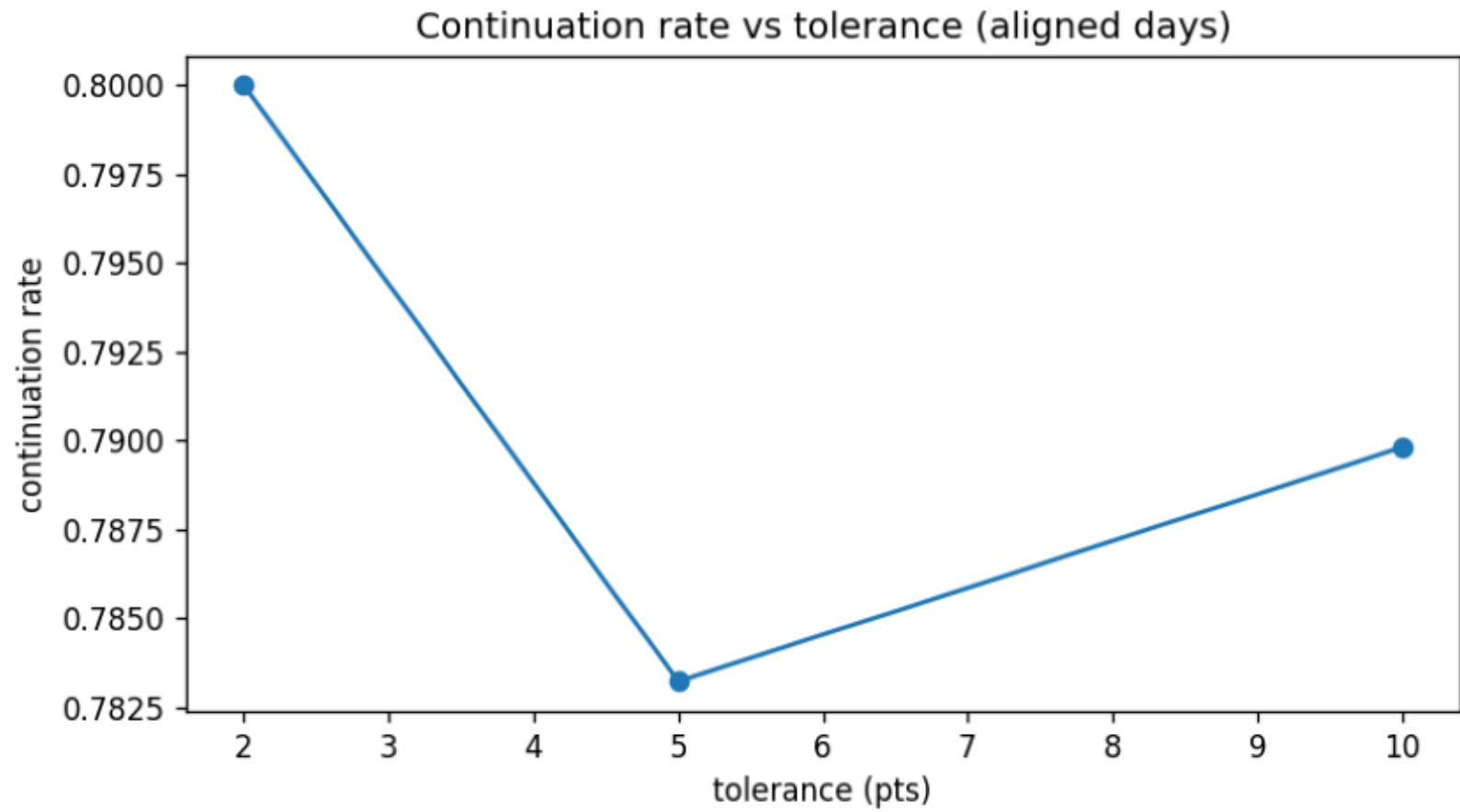


Chart 3 — Top level pair ranking

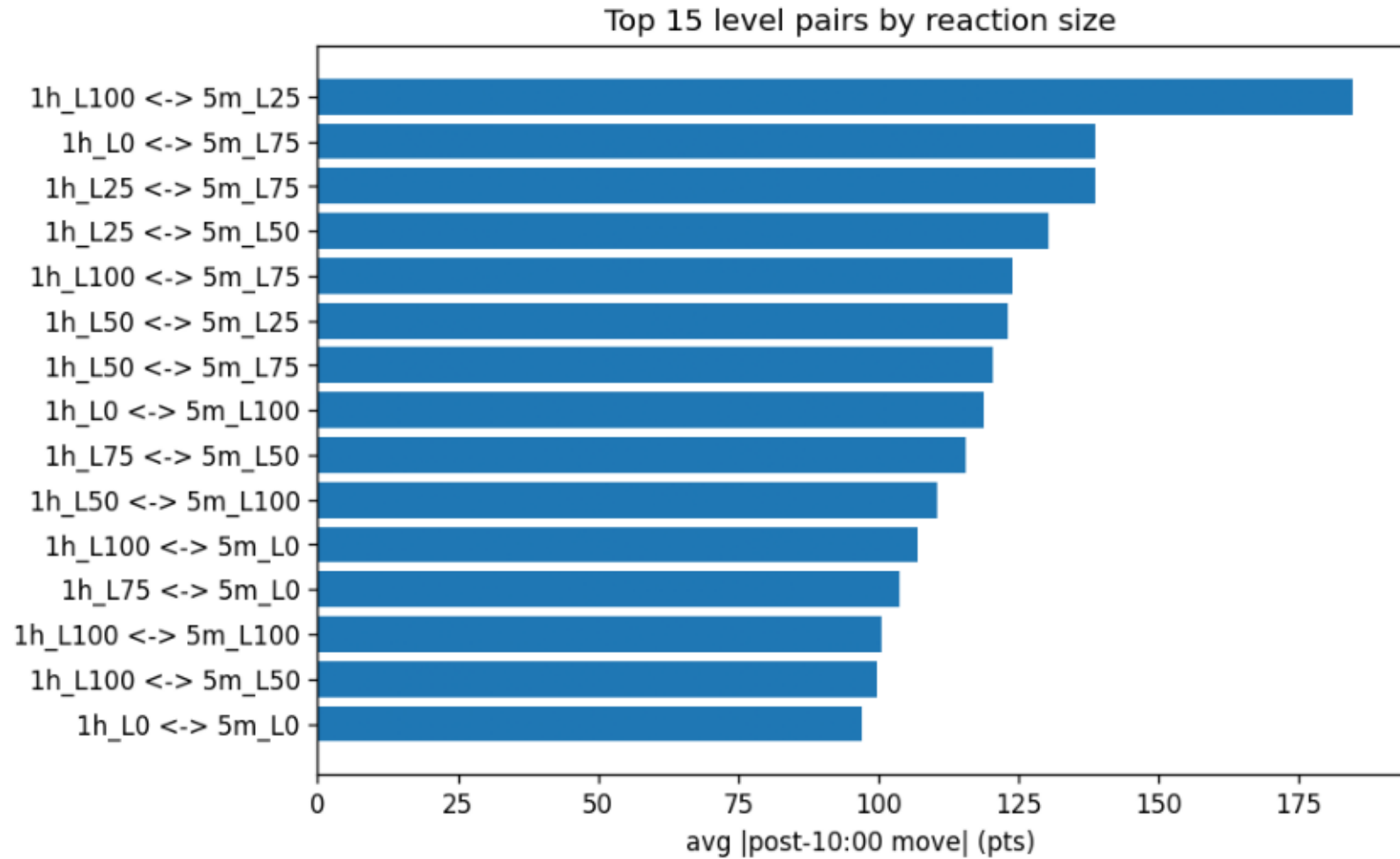


Chart 4 — Alignment distance vs realized move

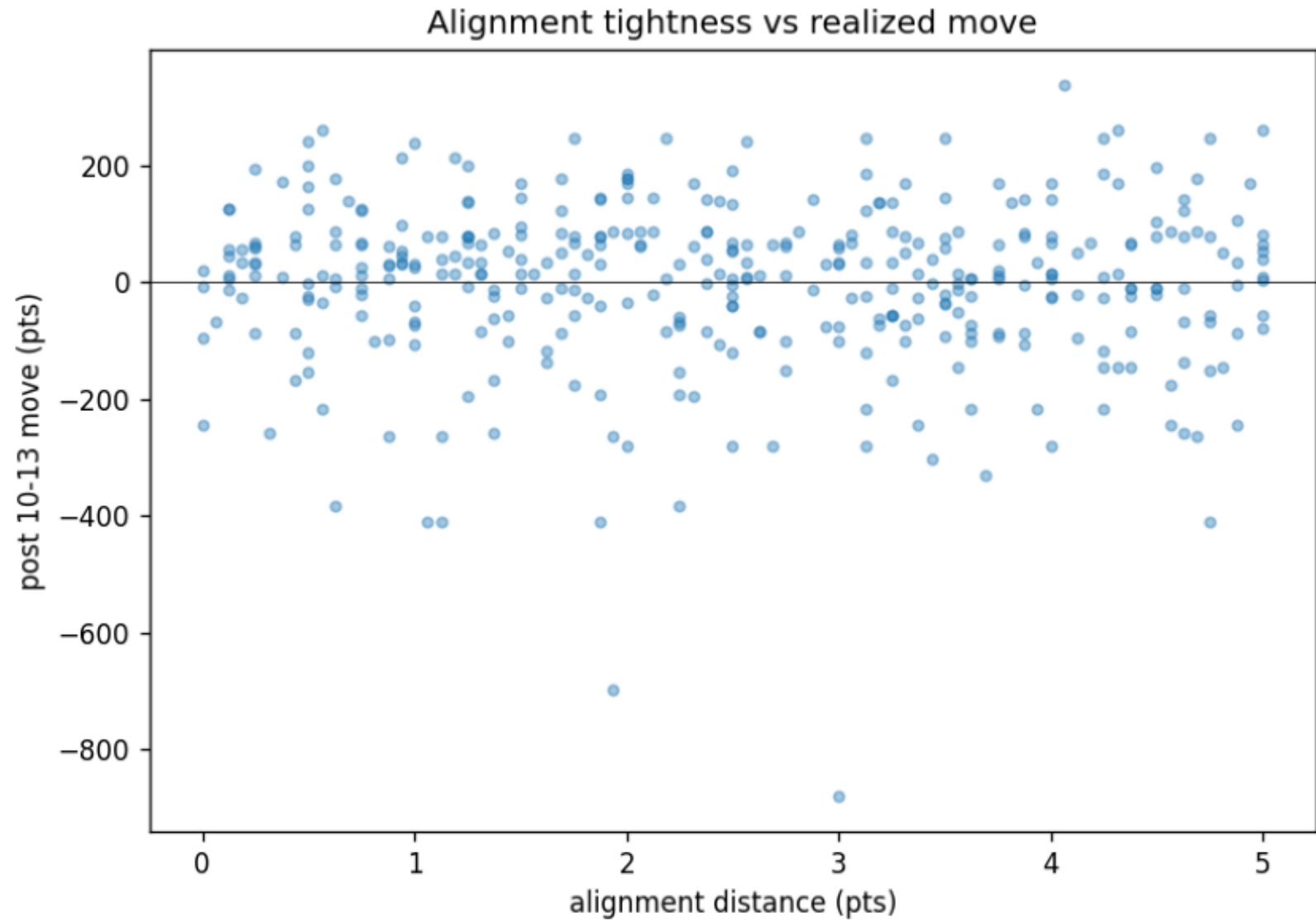
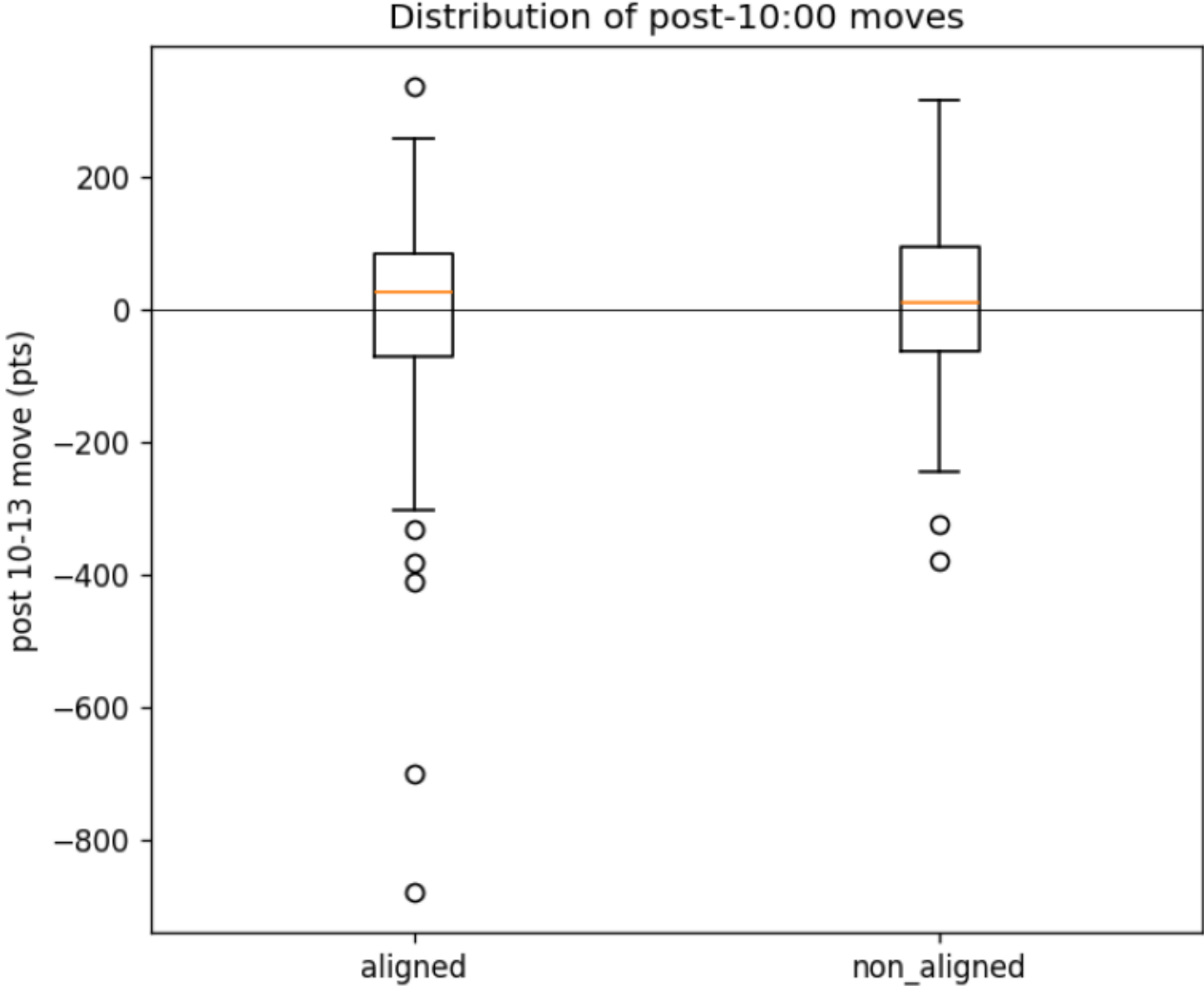


Chart 5 — Distribution of post-10:00 moves



Interpretation

1. HYPOTHESIS: weakly directional but not statistically significant.

Aligned days had higher continuation (78.3% vs 74.5%) and larger average |moves| (115.2 vs 99.9 pts), but every comparison has $p > 0.20$. With $n=253$ sessions we cannot reject the null.

2. RANGE LEVELS RANKED (FIRST TOUCH OF TODAY'S 9AM):

1h_L0 (low wick): cont=82.5% avg|move|=104.6 best

1h_L100 (high wick): cont=82.4% avg|move|=117.7 best

1h_L25: cont=76.4% avg|move|=95.6

1h_L50 (CE): cont=77.8% avg|move|=98.1

1h_L75: cont=74.0% avg|move|=89.8 worst

-> WICKS dominate, NOT body levels. CE is mid-pack here.

3. ALIGNMENT IMPROVEMENT VS NON-ALIGNMENT:

Continuation +3.8pp, |move| +15.3 pts. Real direction, not significant at $p < 0.05$. Effect size suggests ~700 sessions (3 years) needed for confirmation.

4. DOES NON-ALIGNMENT = CHOP? NO.

Chop rate < 1% in BOTH cohorts. Non-aligned days still trended.

The chop framing is REJECTED.

Caveats and next steps

DATA CAVEATS

- One year of MNQ continuous front-month, n=253 sessions.
- 3 days flagged 'degraded' by Databento (2025-09-17, 2025-09-24, 2025-11-28) - kept in the run.
- Continuous-contract roll boundaries can introduce small range distortions on changeover days.

METHOD CAVEATS

- 'Continuation' = today's 9AM hi/lo hit first, post-10:00.
Days that grind sideways without touching either side count as 'inside' (neither cont nor reversal).
- Significance is two-sided z and Welch t with normal-CDF approximation (no scipy dependency). For $df > 30$ this is accurate; for the small subgroups (n=14, n=26) treat the p-values as approximate.
- Reaction window for prior-zone touches: 60 minutes after first touch. Reversal threshold: > tolerance pts.

RECOMMENDED NEXT STEPS

- A. Pull 3-5 years of MNQ 1m (~\$5) and re-run alignment test.
- B. Test 9AM FVGs specifically (3-bar imbalance inside 9:00-9:59)